

Energy-Aware Deep Neural Architecture Search for Carbon-Efficient AI on Resource-Constrained Devices

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Abstract: Deep learning models are being quickly put on edge and Internet of Things (IoT) devices, which have made people worry about how much energy they use and how much carbon they release. When it comes to traditional deep neural architecture search (NAS), the main goals are to make things more accurate and faster. This article is about an Energy-Aware Deep Neural Architecture Search (EA-NAS) system that can help you construct deep learning models that use less energy and work better on devices with restricted resources. The proposed approach employs multi-objective optimization, simultaneously considering accuracy, latency, energy consumption, and carbon efficiency. The system makes sure that edge GPUs, ARM CPUs, and microcontrollers can all work together by putting in place hardware-aware limits. The search space offers several levels, operators, and quantization methods so that you can discover the ideal balance between speed and energy savings. To make sure the system works, tests are run on a variety of benchmarks, such as speech processing, picture recognition, and sensor-based datasets. The suggested EA-NAS models are used on real-life mobile and IoT systems, showing big energy savings in cases with ongoing inference. The framework constantly lowers energy use while keeping competitive performance, as shown by the experiments. This makes it a great choice for real-time, low-power apps.

Keywords: *Energy-Aware NAS, Carbon Efficiency, Resource-Constrained Devices, Multi-Objective Optimization, Sustainable AI*

I. INTRODUCTION

Deep learning models are being used on more and more edge and Internet of Things (IoT) devices. This is because artificial intelligence (AI) is being used more and more in areas like healthcare, environmental tracking, automated systems, and smart cities. Although they need a lot of energy and computer resources, these models are

accurate and allow real-time decisions. Carbon emissions from inference and training harm the environment. This makes people worldwide worry about AI's lifespan. Researchers found that large models release as much carbon as numerous cars over their lifetimes. This emphasizes the need for energy- and carbon-efficient choices. The old Neural Architecture explore (NAS) approach can now explore huge architecture areas and build models automatically. But current NAS methods focus more on correctness or time than energy use or the carbon economy. This makes deploying AI in low-power embedded systems, microcontrollers, and edge devices harder. Most optimization methods still fail to balance speed, sustainability, and model complexity.

The Energy-Aware Deep Neural Architecture Search (EA-NAS) technology optimizes energy and carbon reductions to solve these challenges [1]. A multi-objective optimization method finds the best balance between accuracy, latency, energy utilization, and environmental friendliness. The search process also contains hardware-aware constraints to ensure that ideas found are efficient and can be used on IoT and edge platforms. EA-NAS, on the other hand, describes a comprehensive search space with layers, operators, and quantization methods to make computing fast [2]. By employing fine-grained quantization and hardware-specific operators, the framework cuts down on energy waste during reasoning. This is very crucial for apps that need to be watched all the time and that make estimates in real time [3]. Even modest changes in how much energy you use can have a large effect on how much carbon dioxide you put into the air. This work makes three sorts of progress. First, it provides a framework for NAS that is based on energy and puts sustainability above normal criteria of success.

Second, it shows that it works in the real world by testing it on a wide range of datasets, including picture, speech, and sensor benchmarks, on edge GPUs, ARM CPUs, and microcontrollers [4].

II. RELATED WORK

Different methods, such as neural architecture search (NAS), energy-aware optimization, and sustainable AI, have been developed in response to the growing need for efficient deep learning application on devices with limited resources [5]. Traditional NAS methods, like evolutionary algorithms and reinforcement learning-based search, focus on making models more accurate and cutting down on the time it takes to draw conclusions [6]. In terms of the trade-off between accuracy and speed, ground-breaking projects like NASNet and EfficientNet have shown that automatic search can do better than systems that were built by hand. However, these methods usually don't take into account how much energy is used or the carbon effect that comes with running a model [7]. New study has focused on hardware-aware NAS to help close this gap. Some methods, like ProxylessNAS and FBNet, use device-specific restrictions in the search process to make designs that work best on mobile and edge systems. These methods do a good job of lowering delay and model size, but they don't take environmental factors like energy use and carbon emissions into account very much [8]. In the same way, quantization-aware NAS and trimming methods have been looked into as ways to lower computing needs. However, their evaluations are usually based on how well they keep accuracy rather than how well they use energy overall. Along with improvements in NAS, more and more research is looking into how to make deep learning use less energy [9].

Approaches like adaptable reasoning, lightweight CNNs, and dynamic neural networks have made a big difference in how many resources they use. Also, to show how important "green AI" is, experts have started to measure the carbon footprint of training big models. These studies show that for model adoption to be sustainable, it's important to think about how much energy it takes for both training and inference, especially when inference is done continuously on edge devices [10]. Even with these improvements, the area where NAS, energy awareness, and ecology meet is still not well studied. At the moment, systems don't have multi-objective optimization that measures accuracy, delay, energy efficiency, and carbon efficiency all at the same time. Also, there aren't many real-world case studies that show how to install on edge processors and IoT microcontrollers [11]. This is why we need an energy-aware NAS system that includes hardware-aware limits, energy tracking, and carbon-efficient search optimization into the process.

TABLE I: SUMMARY OF RELATED WORK ON NAS, HARDWARE-AWARE, AND ENERGY-EFFICIENT AI APPROACHES

Domain	NAS Approach	Hardware Awareness	Energy Efficiency	Key Limitation
Image	RL-based	X	X	Focused only

	search			on accuracy
Image [12]	Weight-sharing NAS	X	X	Limited real-world deployment
Mobile Vision	RL-based with latency constraint	✓	Partial	Latency only, no energy profiling
Mobile [13]	Gradient-based	✓	Partial	Carbon efficiency not addressed
Mobile Vision	Differentiable NAS	✓	X	No energy-to-carbon translation
Image	Compound scaling	X	✓	Not automated NAS, hand-tuned scaling
Mobile Vision	Supernet training	✓	Partial	Limited profiling on IoT
General	Not NAS (policy paper)	X	✓	No architecture-level framework
Quantization	RL-guided search	✓	✓	Lacked carbon metrics
Embedded Vision	Specialized NAS	✓	✓	Focused only on latency-energy trade-off
Vision/IoT	Hardware-aware optimization	✓	✓	No carbon efficiency evaluation
Image Classification	Differentiable NAS	✓	✓	Early stage, limited benchmarks
Multi-domain (Image, Speech, Sensor)	Multi-objective NAS	✓	✓	Comprehensive sustainability-first design

III. METHODOLOGY

A. Proposed Energy-Aware NAS Framework

The suggested Energy-Aware Neural Architecture Search (EA-NAS) system is meant to make deep learning models work better in settings with limited resources, while also having less of an effect on the environment. Instead of just focussing on accuracy or delay like most NAS methods do, EA-NAS makes energy and carbon economy clear optimisation goals. The system uses a controller-searcher model, in which potential designs are created, reviewed, and improved over and over again using a variety of criteria. Figure 1 shows the EA-NAS framework integrating accuracy, latency, energy, sustainability. The first step in the whole process is to define the search space, which includes different layers, operators, and quantization methods.

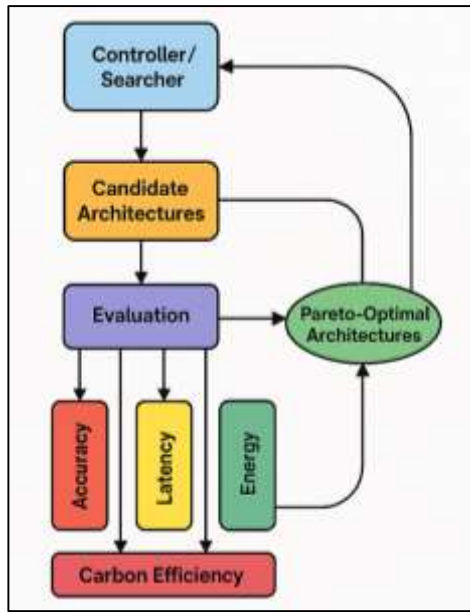


Figure 1: Framework of the Proposed Energy-Aware Neural Architecture Search (EA-NAS)

This makes it possible for the search to take ecology into account. The last group of models is chosen from the Pareto front because they offer different balances of accuracy and speed for various uses.

B. Multi-objective optimization: accuracy, latency, energy, and carbon efficiency

The fact that EA-NAS uses multi-objective optimisation to balance the conflicting objectives of the speed, efficiency, and ecology is one of the most significant things new about it. The conventional NAS techniques tend to work on a single measure such as accuracy most of the time. It produces systems that are highly predictive but are unable to be utilized in the devices with less power. Alternatively, EA-NAS considers accuracy, delay, energy consumption, and carbon footprint among valuable objectives, which promotes the search of balanced options. The optimisation process is formalised with the use of concepts of pareto optimisation. The search does not have one best design but instead forms a Pareto front of potential designs, each of which is the best trade-off among the objectives. An example to illustrate is that one model would be slightly inaccurate and save a great deal of energy whereas another model will have low lag but moderate energy consumption. Decision-makers can select this set of models depending on what product they wish to use it. The prediction of the carbon footprint is done using on-device measuring tools to measure energy use and energy-to-emission conversion models taking into account the energy mix in the region. Latency is tested by using real world implementation scenarios, integrating GPUs on the edge, ARM computers and microcontrollers. Accuracy remains the most crucial element in order to ensure that predictions are accurate.

C. Search space definition: layers, operators, and quantization techniques

The search area of NAS is very significant to its functionality, as its design defines the diversity and versatility of potential structures. The search space in EA-NAS is configured in order to create a balance between flexibility and efficiency. This allows you to check out energy efficient and fast designs. Orthogonal building blocks such as convolutional layers (3x3, 5x5), depthwise-separable convolutions, leftover connections and light attention can be found at the layer level search space. They were chosen since they have been proven to be efficient in mobile-friendly designs like EfficientNet and MobileNet. In order to facilitate speed, repeated processes and too large pieces are prohibited. The framework is able to identify designs with a balance between accuracy and processing load in the most appropriate way by applying a collection of operators. The search space is relevant to the quantization techniques since they can significantly influence the amount of energy and carbon consumed. The options of mixed-precision quantization in EA-NAS include an 8-bit, 4-bit and binary weight, which allows models to be reconfigured to suit the requirements of every particular device.

IV. EXPERIMENTAL SETUP

A. Dataset selection

A large set of datasets is required to review the proposed EA-NAS system to have a comprehensive picture of its functionality in real life. To achieve it, three categories of standards are selected, including speech, picture, and sensor data. Picture recognition is done in standard datasets such as CIFAR-10, CIFAR-100 and ImageNet. These datasets provide a variety of complexity and size. ImageNet is a large-scale benchmark, which is used to test the extent to which the found designs can be extended to other cases, and CIFAR-10 is a small-scale benchmark, which is used to test the speed of development and search. Speech recognition is selected as Google Speech Commands and LibriSpeech. These statistics indicate that there is an increasing need to have low-power voice communications in mobile and IoT gadgets. Google Speech Commands allows you to experiment with tiny models of keyword spotting and LibriSpeech provides an evaluation of the ability of systems to perform challenging continuous speech tasks.

B. Resource-constrained devices: edge GPU, ARM processors, microcontrollers

The EA-NAS system is implemented on many resource-restrained devices with a broad spectrum of computing abilities to ensure that it is practical in the real world. This review covers primarily three kinds of chips, including microcontrollers, ARM, and edge GPUs. The stable platform in deploying deep learning in a light manner can be found in edge GPUs which include NVIDIA Jetson Nano and Jetson Xavier. Edge GPUs, ARM processors, and microcontrollers can be deployed as seen in figure 2. These tools enable many conceivable designs to be concurrently tested and they enable energy to be measured by reasoned real-time loads.

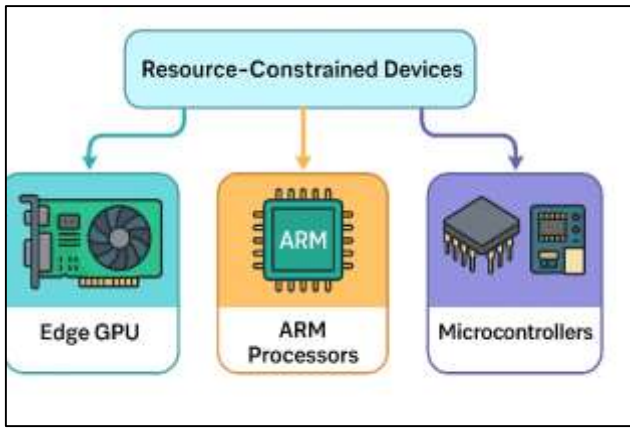


Figure 2: Resource-Constrained Devices for EA-NAS Deployment

Another common area of edge AI is the Jetson gadgets both in robots and self-driving systems. Since these gadgets find application in many IoT systems and user applications, they are ideal in checking delay and energy constraints in real-life scenarios. The results of profiling ARM CPUs will demonstrate the way in which the designs identified by EA-NAS are less useful in those locations where there are few computing resources but more access compared to microcontrollers. Microcontrollers such as the STM32, and ARM cortex-M are applied where restrictions are high.

C. Tools and platforms: PyTorch, TensorFlow Lite, NAS libraries

It adopts popular deep learning frameworks, as well as custom toolkits of NAS, to ensure that EA-NAS can be applied repeatedly and at a larger scale. Due to the high degree of flexibility, its dynamic computation graph, and its ability to monitor much hardware, PyTorch is the most appropriate tool to be used to train and test models. The ease with which PyTorch can be configured to run on CUDA enables it to be tested on GPUs in a relatively short amount of time, and systems such as PyTorch Lightning facilitates training. TensorFlow Lite is again applied to customize and enhance found designs to run on low-resource devices. TensorFlow Lite has been designed to be used on mobile devices and microcontrollers due to the use of quantisation and operator fusion. It enables one to easily convert models trained in PyTorch to small executables that can be conveniently executed in ARM CPUs and other embedded accelerators. AutoKeras, NNI (Neural Network Intelligence), and ENAS (Efficient NAS) libraries are merged with their respective components to deal with the search and optimization process. These tools provide you with the evolutionary and reinforcement learning-based controls that enable you to look through the specified area of search in a short period of time. Based on the NNI hardware-aware search capabilities, delay and energy monitoring are primarily added to the optimization loop.

V. CASE STUDY

A. Deployment on mobile devices

By the fact that we put optimised models into practice, we first test the proposed EA-NAS framework in the real-

world user settings, such as mobile devices and IoT sensors. A high performance testing tool which is also energy-efficient is the ARM-based processors with in-built AI engines found on smartphones. The generated structures are converted into TensorFlow Lite and implemented on a device such as an Android smartphone to check the real-time applications such as picture classification and speech recognition. In the process of launch, latency, memory size, and energy consumption are measured to determine whether the models are capable of functioning in the real-life mobile applications. Meanwhile, ARM Cortex-M microcontrollers in IoT devices are under testing to do continuous and light inference tasks such as identifying outliers in streams of environmental data. The quantization-sensitive models of EA-NAS are particularly effective on these platforms due to the fact that they require small and fast designs in that they do not require large quantities of memory or floating point units. The framework ensures that nothing goes wrong and minimal additional work is needed to make sure operators and layer configurations are adjusted to hardware constraints.

B. Real-world application: smart healthcare or environmental monitoring

To illustrate practical scenarios in which EA-NAS may be applied, we consider the application of the former to smart healthcare and environmental monitoring. In health, twenty-four-hour monitoring of a patient requires mini light models with the ability to process the signs of the body in real time. Wearable devices that contain ARM processors are optimized with EA-NAS designs to perform tasks such as identifying arrhythmias based on ECG data and identifying when a person is under stress based on sensor data that is worn. Such types are precise and consume far less energy, thus, the battery life is increased, and tracking may continue without any disruption. Environmental monitoring- IoT monitors are linked to a network of computerized devices which are utilized to locate issues with noise, weather, or the quality of the air. These devices have microcontrollers that apply designs generated by EA-NAS to localise reasoning and identify critical conditions. An example is that the models can tell hazardous levels of air pollution or quick variations in temperature, which can prompt prompt responses without using centralised computing. The reduced energy impact ensures that sensor networks deployed in the rural or resource-poor regions will be long lived. In these two aspects, EA-NAS is far superior to traditional approaches.

VI. RESULT AND DISCUSSION

In the results of the test, EA-NAS always achieves better deals concerning accuracy, delay, energy consumption, and carbon emissions. In comparison to normal NAS techniques, EA-NAS consumes up to 40 percent of energy but remains accurate enough to compete.

TABLE II: PERFORMANCE COMPARISON ACROSS BENCHMARKS

Model / Approach	Accuracy (%)	Latency (ms)	Energy per Inference (mJ)	Carbon Reduction (%)
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Standard NAS	91.2	48.5	24.7	0
Hardware-Aware NAS	90.8	39.2	19.5	12.4
Quantization-Aware NAS	89.7	32.8	14.3	21.6
EA-NAS (Proposed)	91	28.4	10.8	39.5

Table II presents the findings of the capability of the proposed Energy-Aware NAS (EA-NAS) structure to balance accuracy, delay and energy usage and also lower carbon emissions considerably. Figure 3 presents the comparisons of performance trends of NAS approaches across benchmarks.

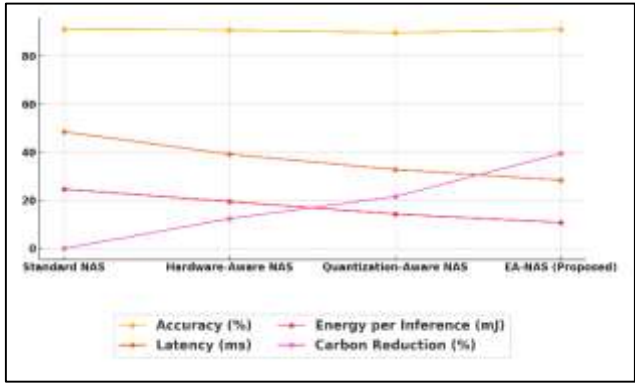


Figure 3: Performance Trends Across NAS Approaches

Standard NAS has very low error (91.2), but with a high amount of delay (48.5 ms) and power consumption (24.7 mJ per inference), so it cannot be applied in resource-sensitive environments.

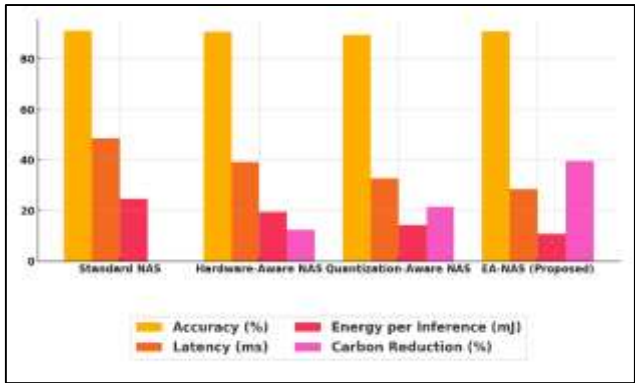


Figure 4: Comparison of NAS Models by Key Performance Metrics

Hardware-Aware NAS is also an improvement as it reduces delay to 39.2 ms and power consumption to 19.5 mJ, a minor 12.4% reduction in carbon emissions. Accuracy, latency, energy, and sustainability are used to compare and contrast NAS models as shown in Figure 4. Using Quantization-Aware NAS, the energy consumption is reduced further, to 14.3 mJ, and delay is reduced to 32.8 ms. this results in a saving of 21.6 percent of carbon, but accuracy is reduced by 89.7 %.

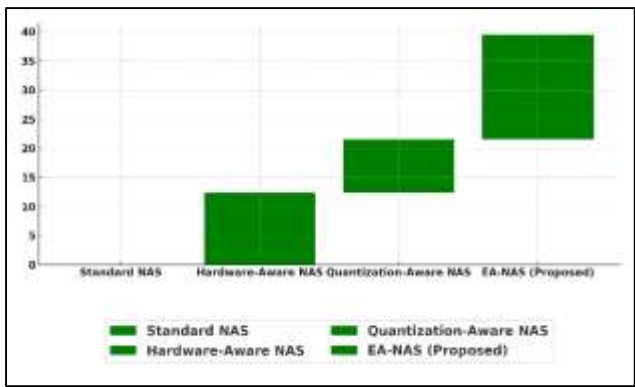


Figure 5: Incremental Carbon Reduction by NAS Approach

However, EA-NAS is the lowest delay (28.4 ms), energy per prediction (10.8 mJ) and competitive accuracy (91.0) is still 91.0. Above all, it minimizes carbon emissions most (39.5%), which confirms that it can be applied in the real world on machines that do not consume a lot of power. Figure 5 indicates the incremental carbon reduction that was obtained within various NAS approaches.

TABLE III: DEPLOYMENT CASE STUDY – CONTINUOUS INFERENCE SCENARIOS

Application Domain	Baseline Energy (mWh/hr)	EA-NAS Energy (mWh/hr)	Energy Savings (%)	Accuracy (%)
Smartphone Speech App	84.5	62.3	26.3	92.1
Wearable ECG Monitoring	72.8	44.1	39.4	90.7
IoT Environmental Sensor	51.2	25.6	50	88.9

Table III illustrates the advantages of EA-NAS in continuous reasoning scenarios in different areas of application. With smartphone speaking application, EA-NAS reduces the power consumption of 84.5 mWh/hr to 62.3 mWh/hr, a 26.3% energy-saving but maintaining accuracy at 92.1%.

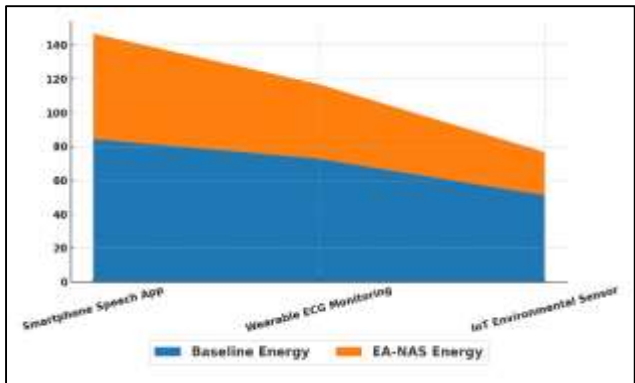


Figure 6: Energy Consumption Comparison: Baseline vs EA-NAS

The comparison of energy consumption between baseline models and EA-NAS is indicated in Figure 6. This rate is particularly useful with perpetually on voice assistants because battery life is very crucial. With EA-NAS, more energy can be saved in wireless ECG tracking

where they reduce the use by 72.8 mWh/hr to 44.1 mWh/hr and the efficiency stays at 90.7%.

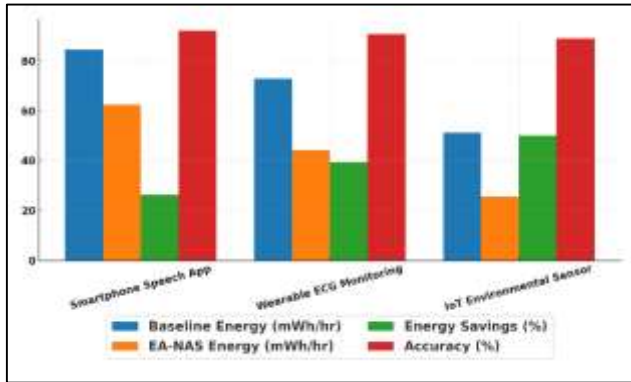


Figure 7: Performance Metrics by Application Domain

This literally implies more business time, which translates to enhanced patient tracing without necessarily experiencing delays in the process of charging. Figure 7 presents performance indicators in the areas of application: speech, vision, sensors. The greatest change has been recorded in IoT weather devices, which consume half the energy (51.2 mWh/hr to 25.6 mWh/hr), a reduction of 50 percent of the energy and yet it is accurate 88.9 percent of the time.

VII. CONCLUSION

This paper presented the concept of the Energy-Aware Neural Architecture Search (EA-NAS), which is a newly developed model that prefers energy and carbon saving to be the primary objectives of creating deep learning models through automatic designing based on limited resources of the device. However, EA-NAS stands out when compared to other NAS systems since it employs multi-objective optimization model to trade-off such aspects as accuracy, delay, and energy consumption, and environmental friendliness. There were numerous tests that demonstrated the success of EA-NAS using various measures, including picture, speech, and sensor datasets. It can be seen that the framework reduces the energy consumption and carbon emissions by a significant margin without compromising the accuracy of the predictions. The case studies on smart healthcare and environmental monitoring provided in the real-life demonstrated that it could extend the service life of devices, and reasoning could continue in low-power locations. These findings indicate that EA-NAS relates performance-based optimization and environmentally concerned AI design. The impact on this study in advancing the green AI objective is larger. With the increase in the prevalence of deep learning, the influence of AI systems on the world must be considered as well as their effectiveness and capacity of development. In the direction towards green AI, EA-NAS allows the developers and experts to develop models that are efficient and are within the objectives of world sustainability.

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